

Computational supplement for Relational clocks and causal Cartan contraction through arity three

GPT-5.6.sol*

Working draft, July 2026

Abstract

This supplement records the claim-to-certificate map, sparse-operation inventory, replay commands and release gates for `09-relational-clocks-berger-d-cartan.tex`. It separates mathematical arguments from machine receipts and keeps the paper fail-closed: a certificate supports only the claim inside its declared phase space and boundary. The working draft is not yet marked `THEOREM_FROZEN`.

The authorship line describes the research archive. Venue-specific human accountability and authorship-policy review remain release questions and are not silently treated as completed by this supplement.

1 Focused claim ledger

The authoritative machine-readable ledger is `../d_quotient_classical/certificates/PAPER_09_BERGER_CLAIM_TABLE.json`. The table below is its human-readable projection.

*OpenAI model. Contributed the research programme, mathematical direction, derivations, machine verification, referee analyses, and manuscript. The project was commissioned and directed by Asger Alstrup Palm (`asger@area9.dk`), who initiated the questions, served as the non-technical orchestrator and corresponding human contact, but claims no technical contribution.

Claim	Mathematical content	Authoritative certificate
P09-C1	Exact smooth non-conformally-flat Berger clock family	POSITIVE_BERGER_CLOCK_BACKGROUND.json
P09-C2	Standard-sign target, timelike phase, positive quartic and energy inequalities	POSITIVE_BERGER_CLOCK_BACKGROUND.json
P09-C3	Internal current, nonzero Q_R and $\Omega_{\text{total}}(\delta, \mathcal{L}_D) = \omega\delta Q_R$	BERGER_CLOCK_REDUCED_CHARGE_SEED.json
P09-C4	Fixed-coupling lapse constraint, compact averaging, $\delta Q_R = 0$ and linear presymplectic nullity	BERGER_FIXED_COUPLING_DELTA_CHARGE.json
P09-C5	Complete cyclic gauge-fixed 54-row unary complex and support-local $54 \rightarrow 26$ contraction	BERGER_GAUGE_FIXED_NONMINIMAL_COMPLETION.json
P09-C6	Advanced/retarded all-row chain contractions, support, adjointness and D -equivariance	BERGER_54_ROW_CAUSAL_GREEN_HOMOTOPY_V2.json
P09-C7	Complete arbitrary-input support-local q_2 , cyclicity, L_∞ identity and D derivation	BERGER_SUPPORT_LOCAL_Q2.json
P09-C8	Complete arbitrary-input support-local q_3 , quartic cyclicity, arity-three identity and $L_D^{(3)} = 0$	BERGER_SUPPORT_LOCAL_Q3.json
P09-C9	Cyclic causal Cartan contraction through arity two	BERGER_CAUSAL_D_CARTAN_V2.json
P09-C10	Closed arity-three source and cyclic C_4 causal primitive on all 54 rows	BERGER_ARITY_THREE_D_CARTAN_FULL_4D.json

All paths in the JSON ledger are relative to `d_quotient_classical/certificates/` and are pinned by SHA-256. The verifier rejects missing files, digest drift, a false required flag, a true forbidden downstream flag, or a manuscript claim identifier absent from both paper sources.

An additional strategic cross-check is pinned separately by `BERGER_Q3_ACTION_SECTOR_CROSSCHECK.json`. It supports the action-to-payload provenance of P09-C8 in two exact sectors but does not replace the complete `BERGER_SUPPORT_LOCAL_Q3.json` certificate.

2 Invariant complex and component realization

The invariant retained complex consists of spatial diffeomorphism ghosts, the dressed symmetric metric, its antifield, and the spatial identity row, with component ranks

$$3 \longrightarrow 10 \longrightarrow 10 \longrightarrow 3.$$

The 54-component gauge-fixed realization adds eight minimal clock, temporal-diffeomorphism and Weyl components and twenty nonminimal components:

$$54 = 26 + 8 + 20 = 26 + 28.$$

The exact degree decomposition is

$$5z^{-1} + 22 + 22z + 5z^2.$$

The retained 26 components are the propagating/constraint endpoint of the support-local contraction; the complementary 28 components form explicit contractible pairs. This separation keeps the theorem invariant under a different ordering of the same gauge-fixed rows.

3 Exact operation inventory

Operation	Rows	Nonzero rows	Sparse PBW terms	Maximum jet order
q_2	54	39	150,305	6
q_3	54	12	5,812,130	8

Both operations use exact coefficients in $\mathbb{Q}(\sqrt{10})$ at the rational Berger fixture. The q_3 portable representation is a strict-JSON manifest plus 54 deterministic gzip-compressed strict-JSON row chunks. Compression is storage only; the operation acts on arbitrary four-dimensional jets and is not a reduced-mode artifact.

The Taylor normalization is

$$Q(\phi) = q_1(\phi) + \frac{1}{2}q_2(\phi, \phi) + \frac{1}{6}q_3(\phi, \phi, \phi) + O(\phi^4),$$

and the invariant producer contract is

$$\langle x_0, q_n(x_1, \dots, x_n) \rangle_{\text{BV}} = \partial_{t_0} \cdots \partial_{t_n} S_{\text{BV}}[\bar{\Phi} + \sum t_i x_i] \big|_{t=0}.$$

The PBW payload is therefore an implementation of a variationally defined operation, not its definition.

The recorded q_3 Tier-2 stages are:

1. action differentiation on the raw minimal complex;
2. canonical clock and gauge-fermion transport to all 54 rows;
3. row-bounded exact replay of graded symmetry, Q^2 , quartic cyclicity and D derivation;
4. portable row-chunk finalization;
5. independent schema, hash, PBW and adjacent-Koszul audit.

The independent consumer audits the frozen operation but does not independently rederive the Bach-clock expansion from the action. The paper states this distinction explicitly.

4 Independent action-to-q3 sector check

The targeted second route writes the homogeneous reduced density independently in

$$h = \hat{h}_{00}, \quad r = R, qquad N = \sqrt{1 - h - 2r}, \quad k = 1 - 2r, qquad \rho = 1 + r,$$

as

$$\mathcal{L}(h, r) = Nk^{3/2} \left[\frac{\alpha_B \bar{C}^2}{8k^2} + \frac{\rho^2 \omega^2}{2N^2} - \frac{\bar{R} \rho^2}{12k} - \frac{\lambda \rho^4}{4} \right].$$

It imports neither the jet classes nor the q_3 producer. In the frozen twice-Euler-density convention, the direct fourth derivatives give

Output	$h^0 r^3$	$h^1 r^2$	$h^2 r^1$	$h^3 r^0$
$\hat{h}_{00}^*$	-27	27/8	81/16	405/128
R^*	-1071/40	-27	27/8	81/16

All sixteen ordered permutations agree with the published row chunks. This is a genuine action-to-payload cross-check in the lapse and dressed radial/Weyl directions. It does not independently regenerate the remaining millions of coefficients, and the certificate forbids that broader reading.

5 Causal domains

The unary maps have the typed domains

$$\Lambda_{54,\pm} : \Gamma_c(\mathcal{C}_{54}) \longrightarrow \Gamma_{J^\pm}(\mathcal{C}_{54})[-1].$$

The local Taylor operations preserve intersections of input supports. The cyclic higher primitives use both time orientations and land in the two-sided causal hull. They are not the causal difference $\Lambda_+ - \Lambda_-$ and are not claimed to be separately retarded or advanced.

6 Cartan sign and support audit

At arity three, the source

$$A_D^{(3)} = [q_3, \iota_D^{(1)}] + [q_2, \iota_D^{(2)}] - L_D^{(3)}$$

has two normalized closure channels. Their exact coefficients are

$$-\frac{1}{2} + \frac{1}{2} = 0, \quad [q_3, D] = 0.$$

The first equality is graded Jacobi; the second is the imported local D -derivation identity with $L_D^{(3)} = 0$.

For cyclicization, the operator output is first paired with the degree-one odd Darboux form. The resulting four-linear tensor uses the Koszul rotation

$$(-1)^{|x_0|(|x_1|+|x_2|+|x_3|)}.$$

The actual degree multiplicities are

$$5z^{-1} + 22 + 22z + 5z^2.$$

The coefficient of z^0 in its fourth power is 978,736. Exhaustion over these actual degree classes gives zero C_4 group-law defects. Exact group algebra arithmetic verifies

$$\left(\frac{1 + \tau + \tau^2 + \tau^3}{4} \right)^2 = \frac{1 + \tau + \tau^2 + \tau^3}{4}.$$

The support claim is

$$\text{supp } \iota_{D,\text{cyc}}^{(3)}(f, g, h) \subseteq J(\text{supp } f \cup \text{supp } g \cup \text{supp } h).$$

The certificate deliberately rejects the stronger statement that this cyclic primitive is separately retarded or advanced.

7 Fast replay and frozen heavy tier

From the repository root for this programme, the focused fast checks are:

```
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_causal_d_cartan_arity_three.py
```

```

PYTHONPATH=. pytest -q \
  d_quotient_classical/backreacted_clock/tests/\
  test_berger_causal_d_cartan_arity_three.py \
  d_quotient_classical/backreacted_clock/tests/\
  test_paper_09_claim_table.py

PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  berger_q3_action_sector_crosscheck.py --check --guards

PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_q3_action_sector_crosscheck.py

```

The complete q_3 derivation is not part of this fast rail. Its recorded Tier-2 receipt is `../d_quotient_classical/certificates/BERGER_SUPPORT_LOCAL_Q3_TIER2_RECEIPT.json`. Recompute that tier only when the action, convention, canonical maps, PBW engine or payload changes; otherwise verify and import its content-addressed output.

8 Fail-closed boundaries

The main theorem depends on the conjunction of the ten claims above. It does not promote any of the following:

- all-orders nonlinear D -Cartan contraction;
- arity-four closure;
- an integrated nonlinear gauge quotient or global complete observable;
- a general-background nonlinear Koszul–Tate export;
- Hadamard or Feynman two-point functions;
- renormalized composites, anomaly cancellation or the quantum master equation;
- a positive graviton Hilbert space or interacting unitarity;
- a boundary, asymptotic, black-hole or scattering charge theorem.

The positive matter clock and the vanishing compact-universe tangent charge must not be transported to those settings without a new phase-space and boundary calculation.

9 Draft-to-freeze checklist

This draft may be promoted to `THEOREM_FROZEN` only when:

1. every pinned certificate and independent verifier passes from a clean committed tree;
2. the release manifest records the source commit, operating system, Python, SymPy, JSON Schema and pytest versions, and distinguishes the fast and heavy rails;

3. the JSON claim ledger, this table and the manuscript theorem have no missing or broadened dependency;
4. the nonlinear team independently imports the frozen q_2 , q_3 and arity-three Cartan hashes and signs off Sections 6–7;
5. the action-to- q_3 strategic cross-check remains exact and is described as partial rather than a full second producer;
6. the abstract, theorem and conclusion retain the classical, fixed-coupling, compact-Berger and arity-three limitations;
7. the compiled main paper and supplement are reproduced from a clean checkout.

The current state is `WRITING_STARTED`; it is intentionally not `THEOREM_FROZEN`.